

Title: Improving generalization of machine learning-identified biomarkers using causal modelling with examples from immune receptor diagnostics

Aim

The primary aim of the paper is to demonstrate how a causal modeling perspective can improve the robustness and generalization of biomarkers identified through machine learning, specifically in the context of high-dimensional molecular data.

The researchers focus on adaptive immune receptor repertoires (AIRRs) as a concrete case study to explore these concepts. The specific objectives of the paper include:

- **Formalizing Challenges:** The authors use causal inference to formally describe and identify challenges that lead to model failure in real-world applications, such as dataset shift, selection bias, confounding, and batch effects.
- **Illustrating Impacts through Simulation:** The paper uses simulations to show how biological and experimental factors influence learned biomarkers and diagnostic performance. This includes demonstrating how changes in confounder distributions or the presence of selection bias can lead to "deceptively good" performance that fails upon deployment.
- **Improving Robustness:** The study argues that causal models help identify stable (invariant) relations between variables that remain consistent across different populations, which is essential for "transportability" or external validity.
- **Guiding Study Design and Processing:** The paper provides practical guidelines and reporting standards for researchers. These aim to help scientists make their modeling assumptions explicit and use best practices - such as importance weighting - to account for known differences between study cohorts and target populations.
- **Establishing a Stringent Framework:** Ultimately, the work seeks to replace "purely associative" machine learning models with models that explicitly consider the data-generating process, ensuring that diagnostics are clinically relevant and generalizable.

Theoretical Framework / Key Concepts

The theoretical framework of the research integrates supervised machine learning with causal inference to improve the robustness of biomarkers, particularly in the high-dimensional domain of adaptive immune receptor repertoires (AIRRs).

Core Causal Inference Framework

- **Structural Causal Models (SCMs):** These models consist of variables and functions that describe the data-generating process, capturing how values are assigned and how they depend on other variables.
- **Causal Graphs (DAGs):** The causal structure is encoded using directed acyclic graphs where nodes represent variables (e.g., age, immune state, AIRR) and directed edges represent direct causal relationships.
- **Do-calculus:** A non-parametric framework described by Judea Pearl used to estimate causal effects from non-experimental data by relying on the assumptions implied by the causal graph.
- **Invariant (Stable) Mechanisms:** Causal models help identify conditional distributions that remain consistent across different populations, which is essential for achieving external validity or transportability.
- **Backdoor Criterion:** A technique for preventing bias by closing all non-causal paths between a cause and an effect while keeping directed causal paths open.

Key Variable Roles in Causal Models

- **Confounders:** Variables that causally affect both the cause and the effect, potentially introducing spurious correlations (e.g., age or sex affecting both the immune state and the AIRR).

- Colliders: Variables influenced by both the cause and the effect; controlling for a collider can inadvertently introduce bias.
- Mediators: Intermediary variables that describe the mechanisms through which a cause influences an effect indirectly.
- Moderators (Effect Modifiers): Variables that change the relationship between a cause and an effect depending on their value (e.g., HLA type in celiac disease).
- Precision Covariates: Variables that influence only the cause or only the effect, which can be used to improve the precision of estimators without introducing bias.

Machine Learning and Dataset Shift Concepts

- Empirical Risk Minimization: The principle used to find a predictive function that minimizes average error over training data.
- Internal vs. External Validity: Internal validity refers to a model's truth within the study population (generalization), while external validity refers to its ability to generalize to new environments or populations (transportability).
- Dataset Shift: A phenomenon where the joint probability distribution of features and labels changes between the source (training) and target (deployment) populations.
 - Covariate Shift: A change in the marginal distribution of the input features.
 - Label Shift: A change in the marginal distribution of the output label, such as disease prevalence.
- Causal vs. Anticausal Prediction:
 - Causal Direction: Predicting the effect from the cause (e.g., AIRR as a consequence of infection); these models are generally more stable under changes in feature distributions.
 - Anticausal Direction: Predicting the cause from the effect (e.g., immune state from AIRR); these models may be less stable as they do not directly model the biological mechanism.

AIRR Domain Concepts

- AIRRs: High-dimensional molecular markers (T-cell and B-cell receptors) that reflect a patient's past and present immune responses.
- V(D)J Recombination: The stochastic process that creates highly diverse immune receptors.
- CDR3 Region: The complementarity-determining region 3, which is the most variable part of the receptor and primarily responsible for antigen recognition.

Methodological Concepts

The methodological concepts in the sources focus on integrating causal inference with machine learning to create robust biomarkers, specifically for adaptive immune receptor repertoires (AIRRs).

Causal Inference Framework

- Structural Causal Models (SCMs): These consist of variables and functions that describe the data-generating process, capturing how values are assigned and their dependencies on other variables.
- Causal Graphs (DAGs): The causal structure is represented using directed acyclic graphs where nodes represent variables (e.g., age, immune state) and directed edges represent direct causal relationships.
- Do-calculus: A non-parametric framework used to estimate causal effects from non-experimental data, relying on the assumptions implied by the causal graph.
- Backdoor Criterion: A technique used to prevent bias by closing all non-causal paths (backdoor paths) between a cause and an effect while keeping the directed causal paths open.
- Variable Role Identification: This involves identifying variables as confounders (affecting both cause and effect), colliders (affected by both), mediators (intermediaries), or precision covariates (affecting only one variable) to determine how to adjust for them.

Evaluation and Validation Concepts

- Internal vs. External Validity: Internal validity (generalisation) refers to a model's truth within the study population, typically assessed via cross-validation or test sets. External validity (transportability) is the ability to generalise results to entirely new environments or populations.
- I.I.D. Assumption: The "independent and identically distributed" assumption, which suggests source and target populations have the same distribution; the sources note this rarely holds in the real world.
- Dataset Shift Detection: Characterising changes in the distribution of variables, including covariate shift (changes in features like age) and label shift (changes in disease prevalence).
- Causal vs. Anticausal Prediction:
 - Causal direction: Predicting the effect from the cause (e.g., immune state leading to AIRR changes). These models are generally more stable under feature distribution changes.
 - Anticausal direction: Predicting the cause from the effect (e.g., using AIRR to predict an infection). These are often less stable as they do not model the biological mechanism.

Mitigation and Correction Strategies

- Invariant Mechanism Discovery: Identifying conditional distributions that remain stable (invariant) across different populations to ensure model robustness.
- Weighting Techniques: Using importance weighting or inverse probability weighting to adjust for known differences between the study cohort and the target population.
- Batch Effect Correction: This includes techniques like regressing out the influence of experimental batches, though the sources note this is more established for gene expression than for AIRR data.
- Data Fusion: Combining multiple datasets sampled under heterogeneous conditions to answer a probabilistic query while dealing with environment changes and confounding.

Simulation and Encoding

- Synthetic Data Generation: Utilizing tools like OLGA (for repertoire simulation), DagSim (for causal graphs), and immuneML (for machine learning analysis) to benchmark models where the ground truth is known.
- AIRR Representation (Encoding): Methods to handle high-dimensional sequences, such as k-mer frequencies (e.g., 3-mers) or summary statistics of receptor characteristics.
- Balanced Error Rate: A performance metric used specifically to account for potentially imbalanced classes in diagnostics.

Assumptions

In the context of identifying biomarkers through machine learning and causal modeling, the sources outline several critical assumptions across theoretical, statistical, and biological domains.

Theoretical and Causal Assumptions

- Causal Structure and Graphs: Causal inference research relies on introducing assumptions about how variables relate to one another. These are typically encoded in a directed acyclic graph (DAG). A fundamental assumption in these graphs is that the absence of an edge between two nodes implies there is no direct causal relationship between them.
- Independent Causal Mechanisms: When predicting in the causal direction (where predictors are the cause and the label is the effect), models rely on the principle that the conditional distribution (the target) is independent of the marginal distribution of the features.
- Model-Specific Roles: The role of a biological variable, such as HLA, is determined by the specific assumptions of the causal model for a particular disease. For example, in viral infections, HLA may be assumed to be a precision covariate, whereas in autoimmunity, it might be assumed to be a confounder or moderator.

Statistical and Machine Learning Assumptions

- The I.I.D. Assumption: Standard machine learning validation assumes that the training and test data are independent and identically distributed (i.i.d.). This implies the source and target populations share the same joint probability distribution, including identical disease prevalence and feature distributions. However, the sources emphasize that this assumption rarely holds in the real world due to dataset shifts.
- Batch Effect Independence: In some diagnostic settings, researchers may assume that batch effects are independent of the labels (the disease state being predicted). Under this assumption, the experimental protocol acts only as a precision covariate that increases variability without introducing systematic bias.
- Internal Validity: Achieving internal validity assumes that the machine learning model has learned the actual source population distribution rather than random noise, which is typically verified through procedures like cross-validation.

Data and Simulation Assumptions

- Simulation as Ground Truth: Because the true rules guiding real-world biological data generation are generally unknown, researchers rely on simulation tools. These tools operate under explicitly defined "ground truth" assumptions - such as specific k-mer signals implanted into sequences - to benchmark the performance of machine learning models.
- Representative Timing: It is often assumed that the study cohort is representative of the target population in terms of the timing of measurement. If training data is collected only after symptoms appear (retrospectively), it may violate the assumption that it accurately reflects the individuals who would actually be tested in a clinical setting.

Methodology

The methodology described in the sources integrates causal inference with supervised machine learning to address the challenges of identifying robust and generalisable biomarkers in high-dimensional molecular data, specifically adaptive immune receptor repertoires (AIRRs).

Causal Modeling Framework

The core methodological innovation is the application of the causal inference framework to formalise and mitigate dataset shifts.

- Structural Causal Models (SCMs) and DAGs: Researchers use directed acyclic graphs (DAGs) to encode the data-generating process, where nodes represent biological and experimental variables (e.g., age, immune state, batch effects) and edges represent direct causal relationships.
- Do-calculus: This non-parametric framework is used to estimate causal effects from non-experimental data by relying on the assumptions implied by the causal graph.
- Backdoor Criterion: This technique is employed to prevent bias by closing non-causal "backdoor paths" between a cause and an effect while keeping directed causal paths open.
- Invariant Mechanism Discovery: The methodology seeks to identify conditional distributions that remain stable (invariant) across different populations, ensuring the machine learning model is robust to shifts in feature or label distributions.

AIRR Diagnostic Development Pipeline

The sources outline a specific pipeline for developing immune-based diagnostics:

1. Cohort Selection: Selecting participants from a source population, often involving "convenience sampling".
2. Sampling and Sequencing: Obtaining approximately 10^5 – 10^6 immune receptors per individual through targeted high-throughput sequencing.

3. AIRR Encoding (Representation): Transforming high-dimensional sequence data into a computationally feasible format, such as k-mer frequencies (e.g., 3-mers), summary statistics of receptor characteristics, or latent representations learned via autoencoders.
4. Machine Learning Training: Fitting models, such as logistic regression with L1 (Lasso) penalty, to learn patterns indicative of the immune state.
5. Directional Prediction: Determining if the diagnostic is predicting in a causal direction (effect from cause, common in autoimmunity) or anticausal direction (cause from effect, common in viral infections), which influences model stability.

Validation and Correction Strategies

- Internal vs. External Validity: Internal validity is assessed via cross-validation or leave-out test sets to ensure the model learns a distribution rather than noise. External validity (transportability) is achieved by discovering mechanisms that generalize to new environments.
- Weighting Methods: Using importance weighting or inverse probability weighting to adjust for known differences between the study cohort and the target population.
- Batch Effect Correction: For transcriptomic data, this involves regressing out the influence of batch indices using linear regression; however, the paper notes this is an open problem for AIRR data, where the authors tested regressing out batch status directly on k-mer representations.

Simulation Methodology

To benchmark these approaches, the researchers utilised several specific software tools to create "ground truth" scenarios:

- OLGA and immuneSIM: Used for simulating realistic AIRRs.
- DagSim: Used to simulate data directly from defined causal graphs.
- immuneML: An ecosystem used to implant synthetic signals into repertoires and perform the machine learning analysis.
- Balanced Error Rate: This was the primary metric used to measure prediction performance in simulations to account for potentially imbalanced classes.

Main Findings

The research provides several key findings regarding the use of causal modeling to improve the generalisability of machine learning (ML) biomarkers, particularly in the high-dimensional domain of adaptive immune receptor repertoires (AIRRs).

1. Superiority of Causal over Associative Models

The authors find that a causal perspective identifies challenges that purely statistical approaches may miss. While associative ML models often fail when applied to new populations, causal modeling improves robustness by identifying stable, invariant relations between variables that remain consistent across different environments.

2. Impact of Confounder Stability (Experiment 1)

Through simulations, the researchers demonstrated that predictive performance is highly dependent on the stability of confounder distributions.

- If a confounder's distribution remains stable between the training (validation) and test populations, performance remains consistent.
- However, if the distribution of a confounder (such as age or sex) changes significantly, diagnostic performance can drop substantially. For instance, a major shift in confounder probability between training and testing led to the balanced error rate nearly doubling (from ~0.26 to ~0.47).

3. Risks of Selection Bias and Spurious Correlations (Experiment 2)

The study finds that selection bias can lead to "deceptively good" performance that does not translate to real-world settings.

- When a study cohort is selected based on factors like a specific hospital, the ML model may learn the signal of the experimental protocol instead of the immune state due to spurious correlations.
- In the absence of a true biological relation, models can still achieve high accuracy by learning these biases, but their performance becomes random when applied to unbiased test data.

4. Stability of Causal vs. Anticausal Prediction

The researchers established that the causal direction of prediction significantly impacts model stability.

- Causal direction: Predicting the effect from the cause (common in autoimmunity) is expected to be more stable under changes to feature distributions ($P(X)$).
- Anticausal direction: Predicting the cause from the effect (common in viral infections) is potentially less stable because the models do not directly capture the underlying biological mechanism.

5. Challenges in High-Dimensional Batch Correction (Experiment 3)

Correcting for batch effects is significantly more complex in AIRR data compared to other molecular markers like transcriptomics.

- While regressing out batch effects in transcriptomic data can result in nearly perfect recovery of biological signals, doing so in the AIRR setting is less effective.
- In AIRR data, trying to correct for batch effects by regressing them out on k-mer representations reduced the error rate but still resulted in higher error than if batch effects were absent. Furthermore, subtle biological signals often remained mixed with spurious signals.

6. Failure of the I.I.D. Assumption

A major conceptual finding is that the "independent and identically distributed" (i.i.d.) assumption - where training and target populations are assumed to share the same distribution - rarely holds in real-world clinical applications. Standard validation (like cross-validation) only ensures internal validity; it does not guarantee the external validity (transportability) required for a biomarker to be clinically useful.

Limitations

The research identifies several limitations and challenges associated with using machine learning (ML) to discover biomarkers, particularly within the high-dimensional domain of adaptive immune receptor repertoires (AIRRs). These limitations range from data collection biases to the inherent complexity of biological systems.

Data and Biological Limitations

- **Sampling Variability:** Observed AIRR data typically reflects only a limited proportion (approximately 10^5 – 10^6 receptors) of a patient's full repertoire, which contains an estimated 10^8 unique receptors.
- **High Dimensionality:** Molecular biomarker data is often extremely high-dimensional, making it difficult to disentangle true biological markers from noise and biases. For AIRRs, representing all 10^{15} potential receptors is computationally unfeasible, requiring researchers to rely on simplified representations like k-mer frequencies or summary statistics.
- **Measurement and Sequencing Errors:** The PCR-based library preparation and sequencing processes are subject to errors that can introduce spurious correlations. These errors are difficult to correct because the "ground truth" (the exact original receptor sequence) is generally unknown.

- **Timing of Measurements:** If study data is collected retrospectively (after a patient has been diagnosed or treated), the resulting biomarker may not be representative of the individuals who would actually be tested in a clinical setting to establish an initial diagnosis,.

Methodological and Statistical Challenges

- **Failure of the I.I.D. Assumption:** Standard ML relies on the assumption that training and test data are independent and identically distributed (i.i.d.), but this rarely holds in real-world clinical applications where dataset shifts are common,.
- **Selection and Convenience Bias:** Data is often collected based on availability (e.g., from specific clinics or genetic backgrounds), which introduces systematic differences that can cause models to fail upon deployment despite strong performance during development,.
- **Anticausal Prediction Instability:** Diagnostics that predict a cause from an effect (anticausal), such as using AIRRs to predict a viral infection, are potentially less stable than causal models because they do not directly model the underlying biological mechanism,.
- **Batch Effect Correction:** While batch correction is established for some molecular data (like transcriptomics), it remains an open problem for AIRR data,. Simulations show that regressing out batch effects in AIRR settings is less effective and can lead to true biological signals being mixed with spurious ones.

Gaps in Biological Knowledge

- **Unknown Causal Rules:** A significant limitation is that the actual rules guiding real-world AIRR data generation are largely unknown, forcing researchers to rely on simulated "ground truth" for benchmarking.
- **Complex Genetic Influences:** The degree to which variables like HLA and genetics leave detectable, predictive marks on the immune repertoire remains a major open question in the field,.
- **Confounder Sensitivity:** Model performance is highly sensitive to the stability of confounder distributions (like age or sex); if these distributions change between the study cohort and the target population, diagnostic performance can drop substantially.

Implication of Practise

The sources provide a framework for biological predictive modelling that prioritises causal reasoning to ensure models remain reliable and transportable across different real-world settings. Rather than focusing solely on accuracy, this approach emphasizes understanding the underlying data-generating process to mitigate the risks of dataset shift and bias.

Causal Design and Theory

The foundation of robust modelling begins with an explicit representation of the data-generating process through the use of directed acyclic graphs (DAGs). These graphs serve as blueprints that make modelling assumptions transparent and help identify which variables require adjustment. A critical distinction must be made between causal prediction (predicting an effect from a cause) and anticausal prediction (predicting a cause from effects). While causal models are generally more stable, anticausal models - such as predicting infection status from immune repertoires - are highly vulnerable to population changes and require more conservative interpretation.

Data Collection and Handling Confounders

Effective model development requires systematic planning during the data collection phase:

- **Confounder Measurement:** All plausible confounders identified in a DAG, such as age, sex, or treatment history, must be recorded consistently. These variables are essential for correcting bias and ensuring model transportability, even if they are not used as primary predictors.
- **Selection Bias:** This should be treated as a structural property of the data rather than a sampling inconvenience. Practitioners must model the mechanism that determines who enters a dataset to avoid predictions driven by sampling artefacts.

- **Batch Effects:** These should not be removed automatically; instead, practitioners should use causal reasoning to determine if a batch adjustment is appropriate, as removing these effects can sometimes inadvertently delete genuine biological signals.

Mitigating Dataset Shift

Practitioners are urged to treat dataset shift as an expected operating condition rather than an exception, as training and deployment distributions rarely remain identical in clinical settings. To address this, the sources recommend:

- **Simulation:** Using synthetic data to stress-test models before deployment allows researchers to see how a model handles confounder shifts, selection bias, or batch effects.
- **Importance Weighting:** When a training cohort differs significantly from the target population (e.g., due to convenience sampling), statistical weighting should be applied to create a more representative pseudo-population.

Validation and Operational Standards

To ensure a model is truly robust, validation must go beyond simple performance metrics on a single dataset:

- **Diverse External Validation:** Models should be intentionally tested against datasets with different confounder distributions, such as varying age groups or geographic locations, to ensure they have learned stable biological relationships.
- **Appropriate Metrics:** For imbalanced diagnostic tasks, practitioners should default to metrics like balanced error rate or sensitivity-specificity trade-offs rather than overall accuracy.
- **Comprehensive Documentation:** Documentation must include the assumed causal structure, identified confounders, and anticipated sources of dataset shift alongside standard performance metrics.

Ultimately, the goal is to prioritise robustness and transportability over apparent accuracy. A model that performs exceptionally well in a lab setting but fails in new populations is considered operationally useless.